One parameter or two?

Predicting erg performance with a power law versus a critical-power model

Verified Concept2 RowErg rankings, season 2026

Generated 2026-11-01

1 Summary

The power law validated in the companion analysis, $t = k\,d^{1.064}$, has one free parameter per athlete (the level k). Given one effort it predicts the time for another distance. We ask whether a model with a second free parameter, above all the critical-power (CP) model, predicts a held-out effort better. Every model is fitted to an athlete's other efforts and asked to predict the time for the distance of the held-out one (leave-one-event-out), on the same 3,993 athlete-seasons as the companion analysis. There are 17,230 predictions common to the six main models.

- **Overall accuracy.** The lowest median absolute error of the predicted time is 2.20%, for PL2 (free exponent). The one-parameter power law (PL1) has 2.25% and ranks 3 of 6.
- **Critical power against the power law.** The classical CP fit is less accurate than PL1 (3.13% against 2.25%; difference 0.88 percentage points [0.80, 0.99]). Fitted by least squares on log time, the same loss as the power laws, CP is less accurate than PL1 (3.02% against 2.25%; difference 0.76 percentage points [0.69, 0.85]).
- **A free exponent.** Letting the exponent vary by athlete (PL2) is more accurate than PL1 (2.20% against 2.25%; difference −0.06 percentage points [−0.12, 0.00]).
- **Where the models differ.** For interpolation the lowest error is PL2 (free exponent) (1.63%); when extrapolating to longer efforts it is PL1 (calibrated exponent) (2.62%) and to shorter efforts PL1 (calibrated exponent) (3.12%).
- **Best practice.** Restricting CP to 2–20 minute efforts gives 3.56% and predicting from the 2 km result alone with PL1 gives 2.52%, against 2.20% for PL1 on all other efforts (on the 6,740 predictions where all eight models exist).

2 Models

All models describe one athlete in one season and are fitted separately for each athlete. Average power on the Concept2 RowErg is $P = 2.8\,(d/t)^3$ (watts, metres, seconds); the constant 2.8 does not affect any prediction because the CP parameters rescale with it.

Table 1: Models compared.

Model	Free parameters	Prediction	Fitting
PL1	1 (level)	$t = k\,d^{1.064}$	geometric mean of the training efforts
PL1 calibrated	1 (level)	$t = k\,d^{\beta_g}$	β_g = pooled exponent of the athlete's sex and age group, estimated on other athletes only (10-fold cross-fitting)
PL1 from 2 km	1 (2 km time)	$t = t_{2000}(d/2000)^{1.064}$	the athlete's 2 km result alone
PL2	2	$\ln t = a + b\ln(d/2000)$	least squares on log time
CP, classical	2 (CP, W′)	$P = \mathrm{CP} + W'/t$, i.e. $\mathrm{CP}\ t^3 + W't^2 = 2.8d^3$	regression of work $W = Pt$ on t, with CP and W′ ≥ 0
CP, log-time fit	2 (CP, W′)	as above	least squares on log time
CP, 2–20 min	2 (CP, W′)	as above	classical fit using only training efforts of 2–20 minutes (at least two needed)
Critical speed	2 (CS, D′)	$d = D' + \mathrm{CS}\,t$	regression of d on t, with D′ ≥ 0

The cubic CP $t^3 + W't^2 = 2.8d^3$ is solved numerically for the time at the held-out distance, so every model answers the same question: *how long would this athlete take for this distance?* The power laws are one- or two-parameter descriptions of the same curve, so PL2 is the like-for-like two-parameter rival of CP. The log-time

CP fit removes a possible disadvantage of the classical regression, which weights long efforts more heavily than the power-law fit does.

3 Evaluation design

Data. The same athlete-seasons as the companion analysis: verified individual RowErg results from season 2026 with at least 3 events spanning a distance ratio of at least 3 (3,993 athlete-seasons, 17,446 results).

Leave-one-event-out. For each athlete-season each event is held out in turn, every model is fitted to the remaining events and predicts the held-out result, giving 17,446 predictions. Two-parameter models therefore have at least two training efforts. For a timed event (4, 30 or 60 minutes) the held-out result is the distance rowed in a fixed time; the model predicts the time for that distance and is compared with the fixed time.

Error. The error is the predicted time divided by the observed time minus one, in percent. We report the median absolute error, its 90th percentile, the median signed error (bias) and the share of predictions within 1% and 2%. Intervals are from a cluster bootstrap (1,000 resamples) that resamples athletes. The comparison with PL1 is paired: the difference of median absolute errors is bootstrapped on the same predictions.

Interpolation or extrapolation. A held-out result is an interpolation if its duration lies between the shortest and longest training efforts, otherwise an extrapolation to shorter or longer efforts. 55% of predictions are interpolations, 22% extrapolate to shorter and 23% to longer efforts.

Common prediction sets. Models are compared on the same predictions. The six main models are compared on every prediction for which all are defined. The best-practice variants (CP restricted to 2–20 minutes, PL1 from 2 km only) need extra data and are compared on the smaller set where all eight models exist.

4 Results

4.1 Overall accuracy

Table 2: Out-of-sample accuracy on 17,230 predictions (median absolute error of the predicted time; difference in percentage points with 95% bootstrap interval; "wins" is the share of predictions, excluding ties, where the model is closer than PL1).

Model	Median error	90th pct	Bias	Within 1%	Within 2%	Difference vs PL1	Wins vs PL1
PL2 (free exponent)	2.20%	8.8%	0.16%	26%	47%	−0.06 [−0.12, 0.00]	50%
PL1 (calibrated exponent)	2.23%	7.4%	0.11%	25%	46%	−0.03 [−0.06, 0.00]	53%
PL1 (exponent 1.064)	2.25%	7.6%	0.09%	24%	45%	–	–
CP (log-time fit)	3.02%	11.5%	−0.29%	19%	36%	0.76 [0.69, 0.85]	39%
CP (classical fit)	3.13%	13.8%	−0.48%	19%	35%	0.88 [0.80, 0.99]	38%
Critical speed	3.51%	20.7%	−0.60%	17%	33%	1.25 [1.16, 1.35]	34%

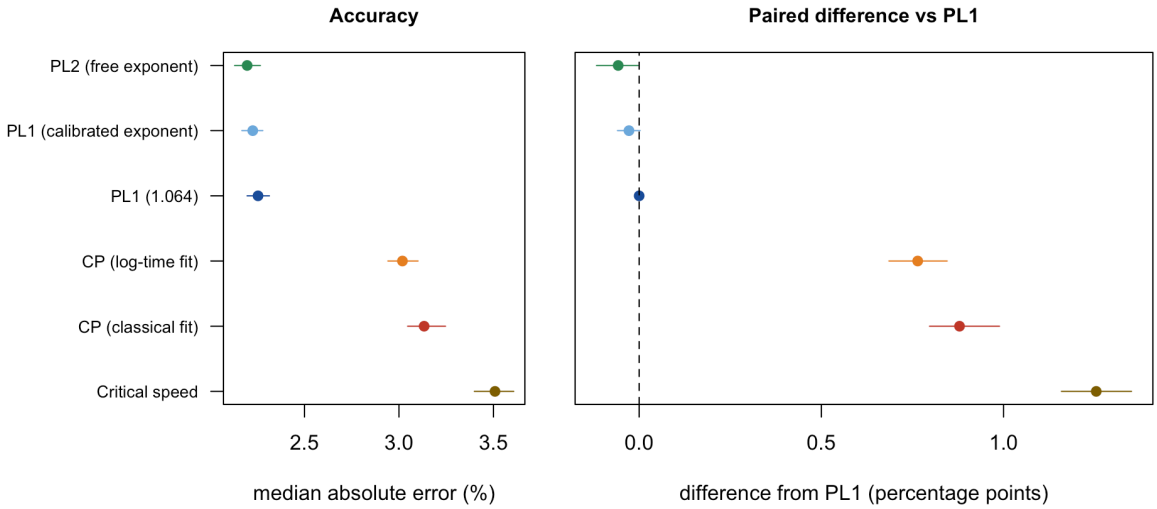


Figure 1: Median absolute error with 95% intervals (left) and paired difference from PL1 (right).

The critical-power model predicts held-out efforts worse than the fixed-exponent power law (Table 2). A power law with a free exponent is more accurate than the fixed 1.064. The two-parameter model with the lowest error is PL2 (free exponent) (2.20%). Figure 2 shows what the fitted curves look like for four athlete-seasons.

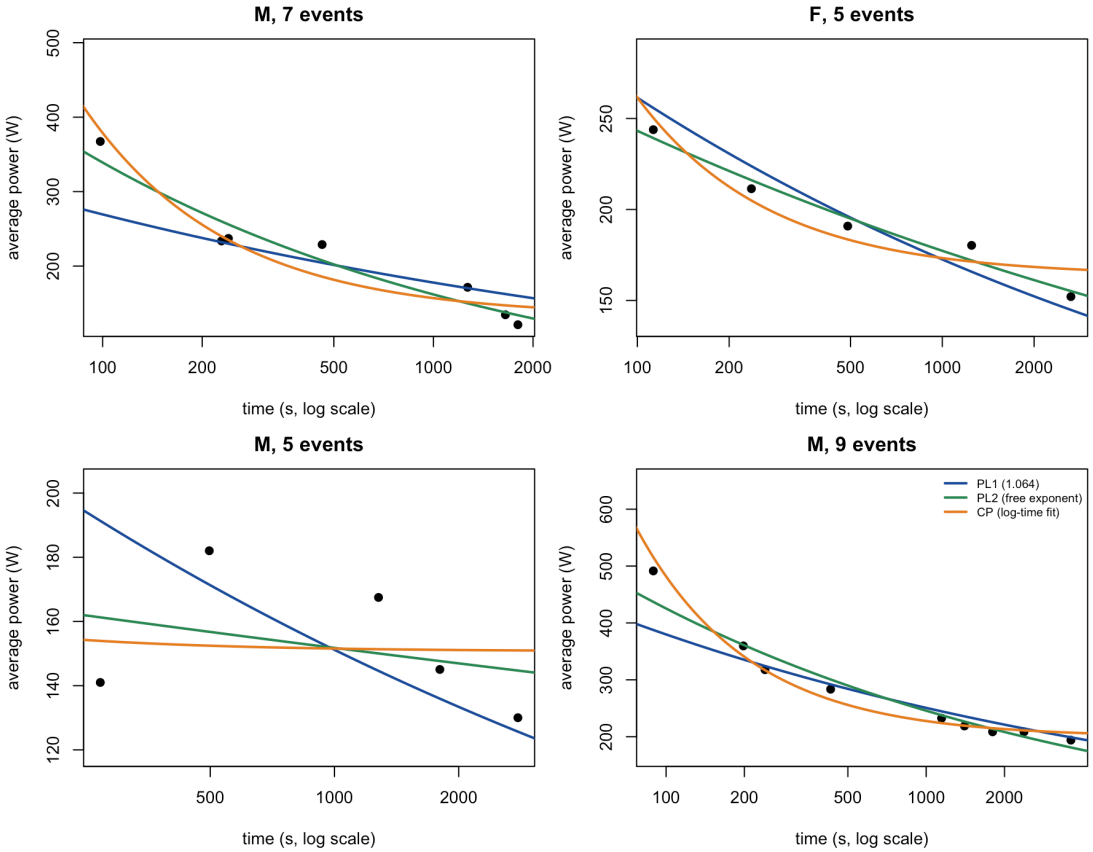


Figure 2: Four randomly chosen athlete-seasons: measured average power against duration with the PL1, PL2 and CP (log-time fit) curves fitted to all of the athlete's events.

4.2 Interpolation and extrapolation

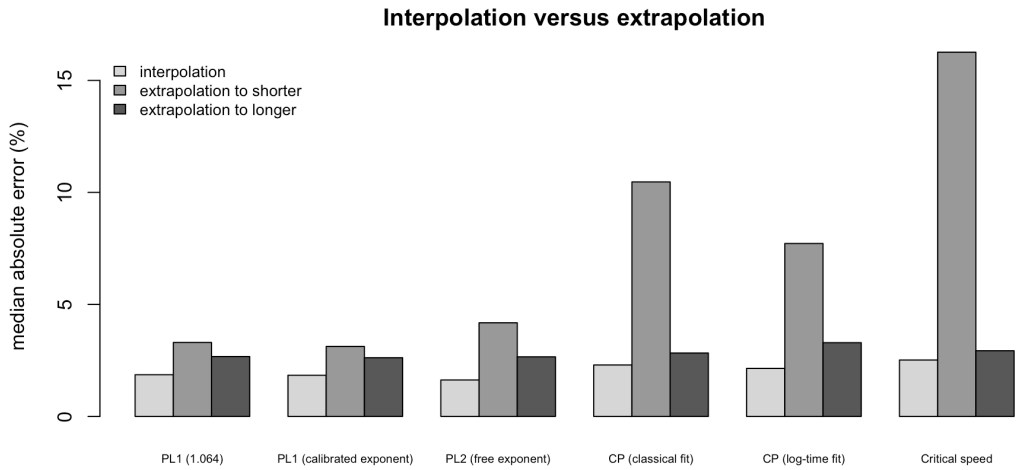


Figure 3: Median absolute error by whether the held-out effort lies inside or outside the range of the training efforts.

Table 3: Median absolute error (%) by position of the held-out effort relative to the training efforts. Bold: lowest in the row.

Held-out effort	n	PL1	PL1cal	PL2	CP	CP log	CS
Interpolation	9,462	1.86	1.84	**1.63**	2.30	2.14	2.52
Extrapolation to shorter	3,775	3.30	**3.12**	4.18	10.47	7.72	16.26
Extrapolation to longer	3,993	2.67	**2.62**	2.66	2.83	3.29	2.93

A second parameter can only help if the extra flexibility is identified by the training data. Table 4 splits the predictions by how widely the training efforts are spread in duration, and by how many there are.

Table 4: Median absolute error (%) by spread of the training efforts' durations. Bold: lowest in the row.

Longest / shortest training effort	n	PL1	PL1cal	PL2	CP	CP log	CS
< 2	576	2.72	**2.62**	8.87	10.48	10.50	11.80
2–4	2,638	2.58	**2.52**	3.27	4.89	4.91	5.61
4–10	4,343	2.28	2.28	**2.24**	3.12	3.05	3.39
≥ 10	9,673	2.15	2.11	**1.86**	2.68	2.59	2.97

Table 5: Median absolute error (%) by number of training efforts. Bold: lowest in the row.

Training efforts	n	PL1	PL1cal	PL2	CP	CP log	CS
2	4,467	2.56	**2.51**	3.09	4.04	4.05	4.53
3	4,054	2.45	**2.37**	2.41	3.23	3.35	3.56
4+	8,709	2.06	2.04	**1.80**	2.71	2.51	2.97

4.3 By held-out event

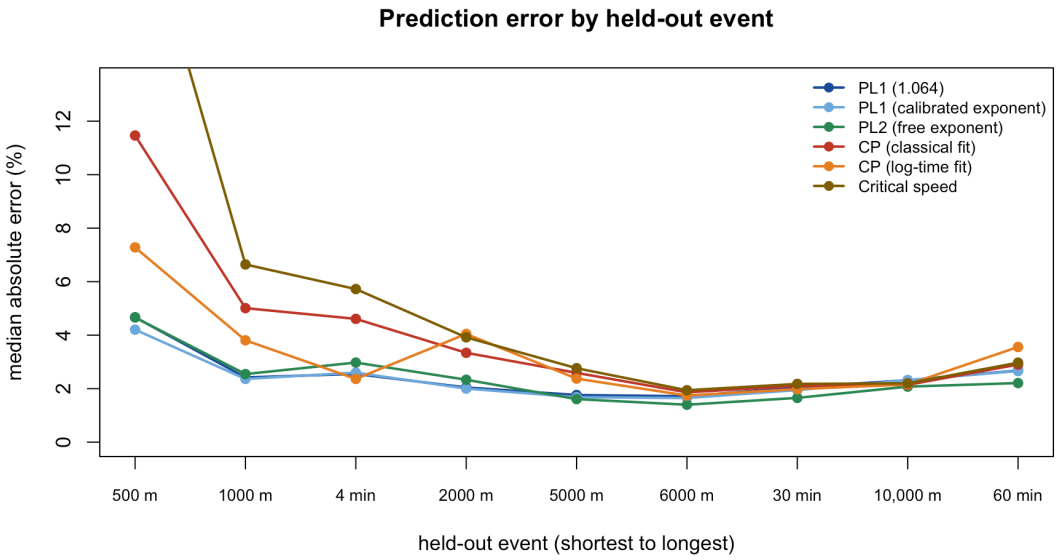


Figure 4: Median absolute error for each held-out event, ordered from shortest to longest typical duration.

Table 6: Median absolute error (%) by held-out event. Bold: lowest in the row.

Held-out event	n	PL1	PL1cal	PL2	CP	CP log	CS
500 m	1,877	4.67	**4.21**	4.66	11.46	7.28	20.92
1000 m	2,337	2.42	**2.36**	2.54	5.01	3.81	6.64
4 min	659	2.55	2.60	2.98	4.61	**2.36**	5.72
2000 m	2,769	2.04	**2.00**	2.33	3.34	4.05	3.92
5000 m	2,549	1.76	1.68	**1.61**	2.59	2.38	2.76
6000 m	1,295	1.72	1.65	**1.40**	1.86	1.74	1.94
30 min	2,190	2.05	1.95	**1.65**	2.10	1.98	2.18
10,000 m	2,260	2.32	2.32	**2.08**	2.14	2.16	2.20
60 min	1,294	2.68	2.66	**2.21**	2.90	3.56	2.97

The lowest-error model differs by event: 500 m: PL1cal, 1000 m: PL1cal, 4 min: CP log, 2000 m: PL1cal, 5000 m: PL2, 6000 m: PL2, 30 min: PL2, 10,000 m: PL2, 60 min: PL2.

4.4 Best-practice variants

Table 7: All eight models on the 6,740 predictions where all are defined (the athlete has a 2 km result among the training efforts and at least two training efforts of 2–20 minutes).

Model	Median error	Bias	Difference vs PL1	Wins vs PL1
PL2 (free exponent)	2.00%	0.30%	−0.21 [−0.30, −0.11]	53%
PL1 (calibrated exponent)	2.20%	0.53%	−0.01 [−0.06, 0.04]	53%
PL1 (exponent 1.064)	2.20%	0.38%	–	–
PL1 (from 2 km only)	2.52%	0.73%	0.31 [0.23, 0.39]	44%
CP (log-time fit)	2.61%	−0.98%	0.40 [0.28, 0.52]	44%
CP (classical fit)	2.96%	−1.04%	0.76 [0.60, 0.88]	39%
Critical speed	3.28%	−1.23%	1.08 [0.92, 1.23]	35%
CP (2–20 min efforts only)	3.56%	−2.79%	1.35 [1.21, 1.49]	35%

On this set CP restricted to 2–20 minute efforts is less accurate than PL1 (3.56% against 2.20%; difference 1.35 percentage points [1.21, 1.49]), and predicting from the 2 km result alone is less accurate than PL1 using all other efforts (2.52% against 2.20%; difference 0.31 percentage points [0.23, 0.39]).

4.5 Fit in the athlete's own data and fitted parameters

The out-of-sample test asks about prediction. Table 8 asks how well each model describes an athlete's efforts in the data it was fitted to, adjusted for the number of parameters (residual standard deviation of log time with degrees of freedom $n - p$).

Table 8: In-sample residual scatter and, among the two-parameter models, the share of athlete-seasons in which each fits best.

Model	Parameters	Median residual SD	Best two-parameter fit
PL1 (exponent 1.064)	1	2.74%	–
PL2 (free exponent)	2	1.89%	66%
CP (classical fit)	2	4.26%	–
CP (log-time fit)	2	2.41%	31%
Critical speed	2	6.09%	3%

Fitting each athlete-season to all of its events, the median exponent of the free power law is 1.067 (interquartile range 1.050–1.085). The median critical power is 170 W and the median W′ is 19.3 kJ (interquartile range 12.9–28.1 kJ). 2.1% of classical CP fits had CP or W′ at the zero boundary.

4.6 One effort is enough for the power law

The power law needs only one effort. Figure 5 shows how well PL1 transfers a single known effort (rows) to every other event (columns) for athletes who have both.

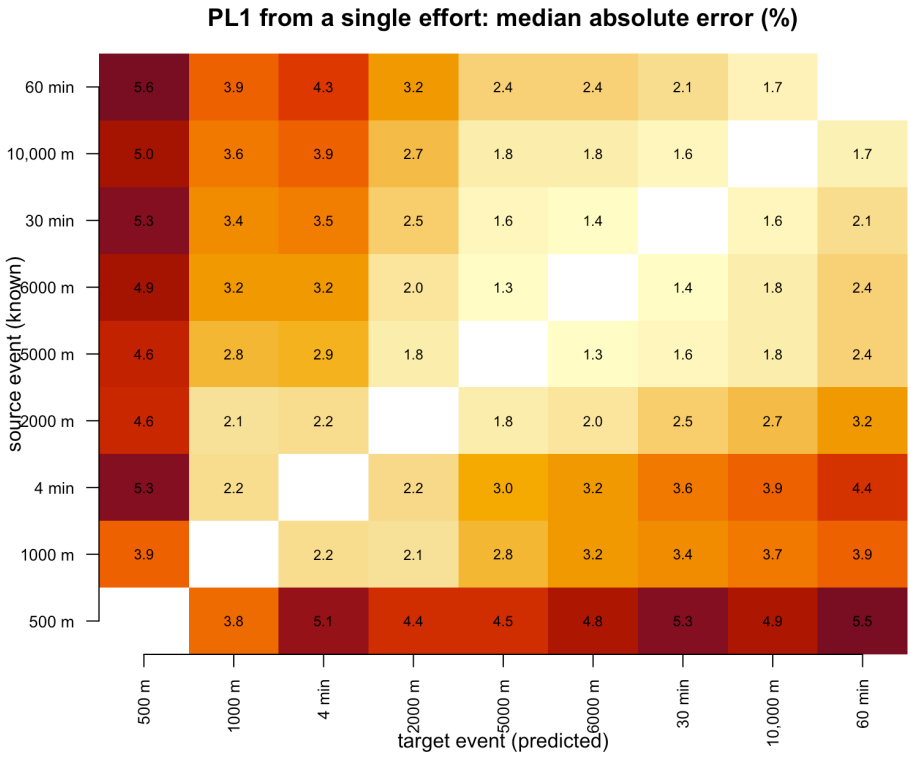


Figure 5: Median absolute error (%) when PL1 (exponent 1.064) predicts the target event from a single source event.

From the 2 km result alone the median absolute error is between 1.8% and 4.6% depending on the target event, and the median signed error runs from −1.2% to 3.6% (positive: predicted too slow).

4.7 Sex and age

Table 9: Median absolute error (%) by sex. Bold: lowest in the row.

Sex	n	PL1	PL1cal	PL2	CP	CP log	CS
Women	3,029	2.41	**2.37**	2.43	3.57	3.35	3.89
Men	14,201	2.21	2.20	**2.15**	3.06	2.96	3.43

Table 10: Median absolute error (%) by age group. Bold: lowest in the row.

Age group	n	PL1	PL1cal	PL2	CP	CP log	CS
≤18	372	2.57	**2.49**	2.82	4.81	4.80	5.19
19–29	658	2.35	**2.14**	2.42	4.39	4.46	4.87
30–39	2,508	2.57	**2.51**	2.62	3.63	3.67	4.18
40–49	4,602	2.33	2.34	**2.30**	3.29	3.14	3.66
50–59	4,556	2.18	2.19	**2.11**	3.09	2.95	3.38
60+	4,534	2.10	2.00	**1.93**	2.60	2.42	2.84

5 Discussion

What the second parameter buys. The critical-power model predicts worse than the fixed-exponent power law. The two models differ in shape as well as parameter count: CP implies a power that levels off at a critical value for long efforts and rises without bound for short ones, while the power law implies a smooth decline across all durations. Letting the power-law exponent vary by athlete also helps, which is consistent with exponents that differ between athletes.

Interpolation and extrapolation. A two-parameter model has more freedom and, with two or three training efforts, little data to pin it down. Table 3 and Table 4 show where that matters: whether the advantage or disadvantage of the two-parameter models sits in interpolation or in extrapolation, and whether it depends on how widely the training efforts are spread. Predictions that extrapolate beyond the training range are the usual

practical need (a 2 km time from short tests, or a 10 km time from a 2 km result) and are where the choice of functional form matters most.

Validity range of CP. The CP model is normally applied to efforts of roughly 2 to 20 minutes, where the severe-intensity domain is assumed; outside that range it is known to misrepresent both very short efforts and very long ones. The events here run from about 1.5 minutes (500 m) to 60 minutes, so the unrestricted CP fits are used well outside their usual range. Table 7 tests the restricted variant, and Table 6 shows how the error depends on the event.

Effort. Both families assume maximal efforts. A single sub-maximal result can distort a two-parameter fit to two or three efforts, either by being absorbed into a distorted curve or by being amplified in extrapolation. The analysis cannot separate this from true model error; the errors reported here include both.

Limitations. The athletes are those with at least three verified events across a threefold range of distance, mostly male masters rowers, from one season. The comparison is on the typical error of one held-out result, not on the best case for any single athlete. The calibrated exponent is cross-fitted by athlete, but it uses the same data source as the evaluation. Performance of CP with more training efforts, with a purpose-designed testing protocol, or with efforts chosen inside its validity range may be better than reported here. Intervals reflect sampling variation only.

6 Conclusion

On 17,230 held-out efforts the model with the lowest median error is PL2 (free exponent) (2.20%), while the one-parameter power law with exponent 1.064 has 2.25%. A critical-power model is less accurate than the one-parameter power law on these data, and a free exponent also improves on the fixed 1.064. The one-parameter model needs only a single effort, which counts in its favour when a second well-spaced maximal effort is not available. The comparison does not say that either model is physiologically correct, only how well each predicts the next effort.

7 Appendix: reproducibility

Run `power_law_analysis.R` first (it prepares the data and defines the athlete-season eligibility rule), then `cp_vs_powerlaw.R`, which sources it and writes `results/cp_*.csv`, `results/cp_key_numbers.json` and `results/fig_cp_*.png`. Compile this report with `typst compile report_cp.typ`. Settings: 1,000 bootstrap resamples, 10-fold cross-fitting of the calibrated exponent, CP efforts restricted to 2–20 minutes in the restricted variant, power constant 2.8. The predictions file contains an anonymous athlete index rather than athlete identifiers.